THREE WORLD MODEL PARADIGMS FOR EMBODIED INTELLIGENCE

Pixel Video Rollouts vs Latent Space Dynamics (JEPA) vs Structured Geometric State Trees

PARADIGM 1: GENERATIVE VIDEO

Pixel-Level Diffusion Rollouts

- Predicts raw RGB video frames $\hat{x}_{t+k}$
- High compute (> 500 ms per step)
- Hallucinates physics & phantom obstacles
- Unusable for 1 kHz safety loops

PARADIGM 2: LATENT WORLD MODEL

Joint Embedding Predictive Architecture

- Predicts abstract latent representations $\hat{z}_{t+k}$
- Focuses on predictable dynamics, ignores noise
- Fast feature prediction (15–30 ms)
- Requires spatial decoder for geometric bounds

PARADIGM 3: STRUCTURED GEOMETRY

Dynamic SE(3) State Estimation

- Predicts metric poses, velocities & covariances
- Deterministic Kalman / Factor Graph update
- Microsecond execution ($\leq$ 100 μs on MCU)
- Directly feeds CBF safety enforcers $h(x) \geq 0$